HopTF: Sequence-Conditioned Associative Retrieval for Hypothesis-Generating Transcription Factor Perturbation Modeling

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Audrey Acken
Columbia University
aa5063@columbia.edu

Peter Driscoll
Columbia University
pvd2112@columbia.edu

Daniel Meyer
Columbia University and New York Genome Center
djm2261@columbia.edu

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Abstract

We introduce HopTF, a framework that combines associative retrieval over genome-wide regulatory context with conditional flow matching to predict perturbation responses across cell states from transcription factor (TF) isoform sequence. TF perturbation responses depend on the molecular properties of the expressed protein isoform and on the regulatory contexts available in each cell state, but many models for predicting perturbation responses do not make the link between protein sequence and regulatory loci explicit. HopTF makes this link explicit through an intermediate representation in which TF sequence variation changes the retrieval distribution over candidate regulatory loci before shaping the predicted shift in cell state. Using TF Atlas overexpression data, we pair isoform-level protein sequence embeddings with genome-derived locus representations and train HopTF to use retrieved context when modeling transport from unperturbed to perturbed single-cell states. We evaluate HopTF through perturbed state prediction and retrieval diagnostics, including sensitivity to sequence variants, shifts in retrieval patterns across loci, and biological support for prioritized regulatory regions. In controlled benchmarks, adding TF sequence embeddings and HopTF retrieval features did not improve perturbed state prediction beyond measured covariates, but variant analyses and a sweep over retrieval temperature show that retrieval remains sequence sensitive and can be tuned from diffuse averaging over many loci to more focused retrieval over a smaller subset. Together, these results position HopTF as a testable framework for regulatory hypothesis generation conditioned on sequence, while underscoring the need for stronger biological validation before treating prioritized regions as causal targets.

1 Introduction

Modeling how transcription factors reshape cellular state is a central problem in systems biology. TFs regulate large gene programs by binding regulatory DNA, interacting with cofactors, and altering transcriptional output. Because perturbing a TF can activate or repress many genes at once, TFs are central to development, directed differentiation, cell-state engineering, and regulatory network inference. The TF Atlas of directed differentiation profiled overexpression responses for more than 3,500 annotated human TF isoforms in H1 human embryonic stem cells using single-cell RNA sequencing [Joung et al., 2023]. This makes TF Atlas a uniquely broad perturbation resource, but it is still fundamentally aggregate and context limited: it records downstream transcriptional responses in one profiled cellular setting, but not the regulatory loci that mediate each response or how those responses should transfer across cell states.

This gap motivates the core modeling question of this project: can a TF isoform sequence be used as a molecular cue to retrieve regulatory contexts from the genome, and can the retrieved context condition prediction of the TF-induced cell-state shift? HopTF addresses this question by combining genome-derived locus representations, protein sequence embeddings, single-cell state representations, associative retrieval, and conditional flow matching. A TF isoform embedding queries a genome-indexed memory bank of regulatory locus embeddings. The resulting retrieval distribution defines a weighted regulatory-context representation, which then conditions a flow model that transports unperturbed cell states toward TF-perturbed states.

The conceptual contribution is not the use of attention or a softmax operation alone. Modern Hopfield networks and transformer attention are mathematically related [Ramsauer et al., 2021, Vaswani et al., 2017], and attention layers are now common across biological deep learning. The distinction is that HopTF assigns the retrieval operation a specific biological role: protein sequence provides the query, genomic loci provide the addressable memory, retrieval produces a regulatory-context hypothesis, and the flow model maps that hypothesis to a predicted cell-state shift. This decomposition makes the retrieval distribution a primary object of analysis rather than an incidental hidden layer.

We therefore frame HopTF as a sequence-conditioned associative retrieval model for hypothesis-generating TF perturbation prediction. The model is trained using aggregate perturbation-response supervision, without labels for which loci mediate each TF response. This induces a weakly supervised multiple-instance structure: candidate genomic loci form a bag of possible regulatory contexts, the TF isoform embedding acts as the query over that bag, and the observed perturbed-cell response is the bag-level target.

Claim boundary. We treat the associative-memory framing as a computational abstraction. We do not claim that the genome is literally a Hopfield memory, nor that high-retrieval loci are validated TF binding sites. Instead, high-retrieval loci are model-prioritized regulatory-context hypotheses. The appropriate biological validation is diagnostic: motif enrichment, accessibility consistency, overlap with external ChIP-seq or perturbation evidence when available, proximity to affected target genes, and robustness to controls based on protein length, ORF length, recovered cell count, and shuffled sequence representations.

Summary of intended contributions.

1. We formulate TF perturbation prediction as sequence-conditioned associative retrieval over a genome-indexed regulatory memory bank.

2. We connect the retrieval module to weakly supervised multiple-instance learning, where locus-level mediators are latent and supervision is available only at the aggregate perturbation-response level.

3. We combine the retrieved regulatory-context representation with a conditional flow field for predicting cell-state transport under TF overexpression.

4. We propose retrieval diagnostics and controls that make the model useful for hypothesis generation rather than opaque prediction of perturbed-cell state.

Figure placeholder: Overview of HopTF

Schematic to create. Left: TF isoform amino-acid sequence is encoded by a protein language model and projected into query space. Middle: genomic loci are encoded by AlphaGenome into a memory bank of keys and values. Cell state optionally modulates retrieval through accessibility or cluster-specific bias. Right: retrieved representation μ_p conditions a flow model that transports control-cell latent states to TF-perturbed latent states. Add a separate panel showing retrieval weights over loci as interpretable activation patterns.

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2 Background and Related Work

2.1 Associative memory, Hopfield retrieval, and attention

Classical Hopfield networks introduced content-addressable associative memory: stored patterns can be retrieved from partial or noisy cues through energy-based dynamics [Hopfield, 1982]. Dense associative memory generalized this idea to higher-capacity energy functions capable of storing and retrieving many more patterns [Krotov and Hopfield, 2016]. Modern Hopfield networks extend associative memory to continuous states and show that the update rule is equivalent to the attention mechanism used in transformers [Ramsauer et al., 2021, Vaswani et al., 2017]. This connection supports a retrieval-centric interpretation of attention: a query retrieves stored patterns through similarity-weighted pooling.

In HopTF, the stored patterns are genomic locus embeddings, the cue is a TF isoform embedding, and the retrieved state is a regulatory-context vector used to predict perturbation response. This perspective has already been useful in biological multiple-instance learning. DeepRC uses transformer-like attention, equivalently modern Hopfield retrieval, to classify immune repertoires from extremely large bags of immune receptor sequences with sparse disease-associated witnesses [Widrich et al., 2020]. The analogy to HopTF is direct but not identical. In immune repertoire classification, the model must identify a small set of disease-relevant sequences from a massive repertoire. In HopTF, the model must identify TF-compatible regulatory contexts from a large set of genomic loci using only aggregate transcriptional response supervision.

2.2 Single-cell perturbation response models

A growing literature models transcriptional responses to genetic or chemical perturbations in single cells. GEARS represents perturbations using gene embeddings and gene-gene relationship graphs to predict outcomes for held-out genetic perturbations [Roohani et al., 2024]. CellOT learns maps between control and perturbed cell-state distributions using optimal transport [Bunne et al., 2023].

scGPT and related single-cell foundation models use transformer-style pretraining over gene expression profiles for cell representation learning and downstream perturbation tasks [Cui et al., 2024]. Recent benchmarking work has emphasized that simple comparisons can be difficult to beat in single-cell perturbation prediction, making controls for dataset structure and rigorous split design essential [Ahlmann-Eltze et al., 2025].

These methods are important comparators, but most do not explicitly represent a TF perturbation by its amino-acid sequence, retrieve regulatory loci, and expose a locus-level activation distribution. HopTF is therefore positioned less as a generic perturbed-state predictor and more as a model for sequence-conditioned hypothesis generation about TF-compatible regulatory contexts.

2.3 Mechanistic and regulatory-network approaches

CellOracle performs in silico TF perturbation by constructing cell-state-specific gene regulatory networks from single-cell multi-omics data and simulating downstream expression changes after TF perturbation [Kamimoto et al., 2023]. It uses accessible promoter and enhancer regions, TF motif scans, and expression-informed active connections to predict how perturbing a regulator shifts target-gene expression and cell-state trajectories. SCENIC+ infers enhancer-driven gene regulatory networks by integrating scATAC-seq, scRNA-seq, motif discovery, enhancer-gene links, and target-gene regression [Bravo González-Blas et al., 2023]. These frameworks provide mechanistic biological interpretability through motifs, regulatory regions, and network structure. However, they generally rely on motif libraries, enhancer-gene linking, and regression or network propagation rather than protein-sequence-conditioned differentiable retrieval.

HopTF is complementary to these approaches. It does not replace motif- or ChIP-based validation. Instead, it produces a learned retrieval distribution over loci that can be compared against motif enrichment, accessibility, target gene annotations, and external binding evidence.

2.4 Protein-aware TF-DNA binding models

TransBind and TFBindFormer are close protein-aware TF-DNA binding baselines because they condition DNA sequence predictions on TF protein representations through cross-attention [Basnet and Cheng, 2026, Liu et al., 2026]. These models are strong comparators when the endpoint is TF-DNA binding prediction on local genomic sequence windows or ChIP-style labels. TransBind is cell-type specific at the output-label level because it is trained on TF-cell-type ChIP-seq labels, but cell type is represented implicitly through the supervised binding labels rather than as an explicit observed regulatory state. TFBindFormer similarly integrates genomic DNA features with TF-specific protein representations for genome-wide TF-DNA binding prediction. HopTF differs by retrieving from a genome-indexed regulatory memory and using the retrieved state to condition a downstream cell-state perturbation model. In other words, HopTF is not a local binding classifier. It is a weakly supervised model of TF sequence, regulatory context, and aggregate perturbation response.

2.5 Sequence-conditioned transport

STRAND is a close conceptual neighbor because it models single-cell perturbation responses using sequence-conditioned transport [Fu et al., 2026]. STRAND represents a perturbation by encoding regulatory DNA sequence at the genomic locus of intervention and uses that sequence representation to parameterize transport from control to perturbed cell states. HopTF differs in the direction of conditioning: the perturbation cue is the TF protein isoform sequence, and the model retrieves from a genome-wide regulatory memory bank before conditioning the flow field.

3 Problem Formulation

Let $x \in \mathbb{R}^G$ denote a single-cell expression vector over G genes. Let p index a TF perturbation, and let s_p denote the amino-acid sequence of the perturbed TF isoform. The training data consist of control cells and TF-perturbed cells. For each TF p, we observe samples from a control distribution P_0 and a perturbed distribution P_p, but we do not observe which genomic loci mediate the perturbation response.

We encode cells into a latent space using a cell encoder E_{cell}, such as SCARF [Liu et al., 2025]:

$$z = E_{\text{cell}}(x) \in \mathbb{R}^{d_z}. \tag{1}$$

The downstream goal is to learn a conditional vector field that transports control-cell latents z_0 toward TF-perturbed latents $z_1 \sim P_p$:

$$\frac{dz_t}{dt} = v_\theta(z_t, t, z_0, \mu_{0p}), \qquad t \in [0, 1], \tag{2}$$

where μ_{0p} is a TF- and cell-state-conditioned perturbation representation produced by associative retrieval over genomic loci.

The key latent variable in the model is the locus activation distribution

$$\alpha_{0pi} = P_\theta(i \mid z_0, s_p), \qquad i = 1, \ldots, N, \tag{3}$$

where N is the number of candidate genomic loci. The model is not trained with labels for α_{0pi}. Instead, α_{0pi} is induced by the perturbation transport objective. This is the weakly supervised multiple-instance structure: the bag is the candidate locus set, the query is the TF isoform sequence, and the bag-level target is the aggregate transcriptional response.

4 Method

4.1 Genome-indexed regulatory memory

We partition the genome into N candidate loci $\{g_i\}_{i=1}^N$. A locus may be defined as a fixed-width genomic window, an accessible peak, a promoter/enhancer candidate, or a gene-linked regulatory region. In the base model, each locus is encoded using a frozen genome foundation model:

$$h_i^{\text{AG}} = E_{\text{AG}}(g_i) \in \mathbb{R}^{d_{AG}}, \tag{4}$$

where E_{AG} denotes AlphaGenome. AlphaGenome provides a regulatory sequence prior because it predicts functional genomic tracks across modalities including expression, accessibility, histone marks, TF binding, chromatin contacts, and splicing-related outputs [Avsec et al., 2026].

HopTF constructs a task-adapted key-value memory from these pretrained locus embeddings:

$$k_i = W_K h_i^{\text{AG}}, \tag{5}$$

$$v_i = W_V h_i^{\text{AG}}. \tag{6}$$

Here k_i is used for TF-locus compatibility scoring, while v_i is the representation retrieved and pooled by the model. A tied-memory variant sets $W_K = W_V$ or directly uses $k_i = v_i$. An untied variant allows the scoring representation and retrieved representation to differ.

Placeholder. Implementation choice to finalize: define the exact locus unit. Options: fixed-width windows, ATAC peaks, promoter/enhancer candidates, gene-linked windows, or AlphaGenome context blocks. State final N, window length L, genome build, overlap/stride, and whether coding regions are excluded or treated separately.

4.2 TF protein sequence query

For each TF isoform p, we compute a protein language model embedding from its amino-acid sequence:

$$e_p = E_{\text{prot}}(s_p), \tag{7}$$

where E_{prot} is ESM-C or another protein language model. A learned projection maps this embedding into the associative retrieval space:

$$q_p = W_Q e_p \in \mathbb{R}^d. \tag{8}$$

This query represents the sequence-derived molecular identity of the perturbed TF isoform. Unlike perturbation-ID embeddings, this representation can in principle distinguish isoforms, sequence variants, domain truncations, or mutants if their protein sequences are provided.

In the current implementation, ESM-C [ESM Team, 2024] uses the 600M-parameter model with mean pooling over non-special tokens, producing a frozen 1152-dimensional isoform embedding. Experiment 1 also tested ESM-DBP, a DBP-specialized ESM2-style 650M model from ESM-DBP [Zeng et al., 2024], with mean non-special-token pooling into a frozen 1280-dimensional embedding. The ESM-DBP run embedded 3264 of 3548 vocabulary entries; 284 entries had no local amino-acid sequence and 139 sequences were truncated to the 1022-token model limit. Missing-sequence rows are retained in the aligned vocabulary with a mask, so downstream baselines can distinguish unavailable sequence information from real zero-valued embeddings.

4.3 Cell-state modulation of retrieval

The retrieval distribution may be global, cluster-specific, or cell-state-specific. The most conservative base model uses no cell-state-specific mask:

$$r_{pi} = \beta q_p^\top k_i. \tag{9}$$

This retrieves TF-compatible loci from a genome-wide memory bank.

A cell-state-conditioned version adds a retrieval bias or mask derived from the cell latent z_0:

$$r_{0pi} = \beta q_p^\top k_i + b_\eta(z_0, h_i^{\text{AG}}), \tag{10}$$

where b_η may represent a learned cell-state/locus compatibility score, a cluster-specific accessibility prior, or a log-accessibility mask. If $m_i(z_0) \in \{0, 1\}$ denotes a hard accessibility mask, this can be written as

$$r_{0pi} = \begin{cases} \beta q_p^\top k_i, & m_i(z_0) = 1, \\ -\infty, & m_i(z_0) = 0. \end{cases} \tag{11}$$

In practice, hard cell-specific masks may be unstable or unrealistic if accessibility is not observed for every target cell. A cluster-specific or cell-type-specific accessibility bias may be a more robust intermediate design:

$$r_{cpi} = \beta q_p^\top k_i + \lambda A_{ci}, \tag{12}$$

where c is a cluster in SCARF latent space and A_{ci} is the average accessibility or availability prior for locus i in cluster c.

Placeholder. Implementation choice to finalize: specify whether the base experiment uses no mask, cluster-specific accessibility bias, or cell-specific accessibility estimates. If accessibility is predicted from SCARF, state how predicted accessibility is calibrated and mapped onto AlphaGenome loci.

4.4 Associative retrieval over genomic loci

Given retrieval scores r_{0pi}, HopTF computes a soft activation pattern over loci:

$$\alpha_{0pi} = \frac{\exp(r_{0pi})}{\sum_{j=1}^{N} \exp(r_{0pj})}. \tag{13}$$

The retrieved regulatory-context representation is

$$\mu_{0p} = \sum_{i=1}^{N} \alpha_{0pi} v_i. \tag{14}$$

In matrix form, with $K \in \mathbb{R}^{N \times d}$ and $V \in \mathbb{R}^{N \times d_v}$, the unmasked version is

$$\alpha_p = \mathrm{softmax}(\beta K q_p), \qquad \mu_p = V^\top \alpha_p. \tag{15}$$

If values are tied to keys, $V = K$, recovering the simpler expression

$$\mu_p = K^\top \mathrm{softmax}(\beta K q_p). \tag{16}$$

This is the associative-memory component of HopTF. Genomic loci are stored patterns, the TF embedding is the retrieval cue, and μ_{0p} is the retrieved regulatory-context prototype used for perturbation prediction.

The temperature parameter β controls retrieval sharpness. Low β yields diffuse averaging across many loci, while high β concentrates mass on a smaller set of loci. We treat β as an inverse-temperature parameter. It may be biologically suggestive of binding selectivity or dosage-dependent sharpness, but we do not assume a direct quantitative equivalence to TF concentration without dose-calibrated data.

4.5 Conditional flow field for perturbation response

The retrieved representation μ_{0p} conditions a time-dependent vector field in cell-state latent space:

$$\frac{dz_t}{dt} = v_\theta(z_t, t, z_0, \mu_{0p}). \tag{17}$$

During training, we use conditional flow matching with minibatch optimal transport couplings [Tong et al., 2024a,b]. For a TF perturbation p, let (z_0, z_1) be a paired sample from a minibatch OT coupling between control latents and perturbed latents. For $t \sim \mathrm{Uniform}(0, 1)$, define a linear interpolation

$$z_t = (1 - t) z_0 + t z_1, \tag{18}$$

with target velocity

$$u_t = z_1 - z_0. \tag{19}$$

The flow matching objective is

$$\mathcal{L}_{\mathrm{CFM}} = \mathbb{E}_{p,(z_0,z_1),t} \left[\| v_\theta(z_t, t, z_0, \mu_{0p}) - (z_1 - z_0) \|_2^2 \right]. \tag{20}$$

At inference, given a control cell x_0, we compute $z_0 = E_{\mathrm{cell}}(x_0)$, retrieve μ_{0p}, and integrate the learned ODE from $t = 0$ to $t = 1$ to obtain a predicted perturbed latent $\hat{z}_1$. This latent can be decoded back to expression space if a decoder is available.

Placeholder. Implementation choice to finalize: specify whether the model predicts final perturbed-cell latent states, latent velocity fields, decoded gene expression, or all three. State whether SCARF is frozen, whether a decoder is used, and how the latent predictions are mapped to gene-level evaluation metrics.

4.6 End-to-end differentiability and weak supervision

The retrieval module is differentiable from retrieval scores through the downstream flow objective. Therefore, even though the model does not observe locus-level labels, gradients from the perturbation-response loss can shape the TF query projection, locus key/value projections, cell-state retrieval bias, and downstream flow model:

$$\mathcal{L}_{\mathrm{CFM}} \to v_\theta \to \mu_{0p} \to \alpha_{0pi} \to (q_p, k_i, v_i, b_\eta). \tag{21}$$

If AlphaGenome, ESM-C, or SCARF are frozen, gradients do not update those foundation model parameters. The base framing is therefore not full fine-tuning of AlphaGenome. Instead, HopTF learns a task-adapted associative regulatory memory initialized by AlphaGenome-derived locus embeddings. A natural extension is to allow adapter or residual updates to the key/value memory slots:

$$k_i = W_K h_i^{\mathrm{AG}} + \Delta k_i, \tag{22}$$

$$v_i = W_V h_i^{\mathrm{AG}} + \Delta v_i, \tag{23}$$

with regularization on Δk_i and Δv_i to preserve the pretrained regulatory prior.

Placeholder. Implementation choice to finalize: specify which components are trained, frozen, or adapter-tuned. If residual memory updates are used, report the regularization strength and whether retrieval diagnostics are robust to these updates.

4.7 Retrieval diagnostics

Because the retrieval distribution is an object of interpretation, not just a hidden layer, we evaluate it directly. For each cell/TF pair, define retrieval entropy

$$H(\alpha_{0p}) = -\sum_i \alpha_{0pi} \log \alpha_{0pi} \tag{24}$$

and effective number of retrieved loci

$$N_{\mathrm{eff}}(\alpha_{0p}) = \exp(H(\alpha_{0p})). \tag{25}$$

We also compute top-k mass

$$M_k(\alpha_{0p}) = \sum_{i \in \mathrm{TopK}(\alpha_{0p})} \alpha_{0pi}. \tag{26}$$

These metrics quantify whether retrieval is diffuse, subset-like, or collapsed. Biological diagnostics include motif enrichment among top-retrieved loci, accessibility consistency, overlap with external ChIP-seq peaks when available, and proximity or linkage to genes with altered expression after TF overexpression.

5 Experimental Design

5.1 Datasets

Primary perturbation data. The main training data are planned to come from the TF Atlas of directed differentiation, which profiles TF isoform overexpression in H1 hESCs at single-cell resolution [Joung et al., 2023]. Each perturbation is associated with a TF isoform sequence and a set of observed perturbed cells. Control cells define the source distribution for flow matching.

The local SCARF-backed TF Atlas subset currently staged for follow-up analysis contains 671,453 cells by 37,528 genes, with 3,266 unique perturbation labels in the cell metadata. It includes 1,000 GFP control cells and 1,000 mCherry control cells. The perturbation labels in the h5ad exactly match the HopTF perturbation metadata used for the controlled sequence-feature analyses.

> **Placeholder.** Fill in remaining preprocessing details: quality-control thresholds, gene filtering, train/validation/test splits, and whether perturbations are split by TF family, held-out isoform, held-out domain, or random TF.

Genomic locus data. Candidate loci are represented by AlphaGenome-derived embeddings. Loci may be defined by fixed genomic windows, accessible peaks, promoter/enhancer annotations, or gene-linked windows.

> **Placeholder.** Fill in: genome build, locus definition, window size, stride, number of loci N, AlphaGenome embedding extraction procedure, whether embeddings are pooled across tracks/layers, and whether locus values include annotations beyond AlphaGenome.

Cell-state representation. Cells are embedded into a latent space using SCARF or another single-cell encoder. In cell-contextualized retrieval variants, cell state may define an accessibility prior or cluster identity.

The staged SCARF file stores RNA-only cell embeddings in `obsm["X_scarf"]` with latent dimension 512. The file provenance records species hg38, counts in `layers["counts"]`, and model genes used: 17,854. The shared cluster path is `/gpfs/commons/groups/knowles_lab/dmeyer/hoptf/data/processed/TFAtlas_subsample_raw_csr_scarf.h5ad`. The May 17 run constructed perturbation-level response vectors as mean TF-perturbed SCARF latent minus the mean control SCARF latent, using the GFP and mCherry controls available in the h5ad. The staged file contains 671,453 cells, 3,266 perturbation labels, and 2,000 control cells; all perturbation labels match the HopTF perturbation metadata used for the sequence and variant analyses.

> **Placeholder.** Fill in final preprocessing details for the camera-ready draft: exact minimum recovered-cell threshold, whether GFP and mCherry controls are pooled or batch-matched in the final run, and whether response vectors are standardized by control-cell latent variance.

5.2 Evaluation splits

We propose several levels of generalization:

1. **Random held-out cells within seen TFs:** tests whether the flow model fits observed perturbation distributions.
2. **Held-out TF isoforms:** tests sequence-conditioned generalization to TFs not seen in training.
3. **Held-out TF families or domains:** tests harder extrapolation across DNA-binding families.
4. **Held-out cell-state clusters:** if multiple cell states are used, tests whether retrieval and transport generalize across cellular context.

Placeholder. Choose final split strategy. For a class report, include at least one perturbation-level split rather than only a random cell-level split, because random cell splits can overstate performance.

5.3 Comparisons and controls

The following comparisons are intended to separate biological signal from architecture and dataset structure:

Table 1: Planned comparisons and their purpose.

Comparison	Description	Question tested
Control / no-perturbation	Predict no movement from control latent.	Is any perturbation modeling learned?
Average TF effect	Use the mean perturbation shift across training TFs.	Are gains TF-specific or just average response?
TF ID embedding	Replace protein sequence with a learned TF identity embedding.	Does protein sequence add value?
ESM-only flow	Condition flow on TF protein embedding without genomic memory retrieval.	Does locus retrieval add value beyond protein sequence?
Random / shuffled memory	Replace AlphaGenome loci with random or shuffled locus embeddings.	Does regulatory memory carry useful signal?
Shuffled TF sequence	Query memory using shuffled or mismatched TF embeddings.	Are retrieval weights TF-specific?
No cell-state bias	Remove cell-state or cluster-specific accessibility bias.	Does cell-contextualization matter?
Length / ORF / cell-count inputs	Predict from sequence length, ORF length, and perturbation cell count.	How much can these measured quantities explain?

Placeholder. Insert final implemented comparisons. Mark any planned but not completed comparisons clearly.

5.4 Perturbed-cell state prediction metrics

We evaluate perturbation prediction in latent and gene-expression space:

- latent MSE or cosine distance between predicted and observed perturbed latents;
- Pearson/Spearman correlation between predicted and observed mean expression shifts;
- correlation restricted to differentially expressed genes;
- distributional metrics such as MMD or Wasserstein distance between predicted and observed perturbed-cell distributions;
- top-k differential-expression recovery or direction accuracy.

Placeholder. Define final primary metric. Suggested: use one main metric for perturbed-cell state prediction and separate diagnostics for retrieval; avoid overloading the paper with too many metrics.

5.5 Retrieval and biological validation metrics

We evaluate the locus activation distribution independently of perturbed-cell state prediction:

– retrieval entropy $H(\alpha)$, effective number of loci N_{eff}, and top-k mass;

– β sensitivity curves showing diffuse, subset, and concentrated retrieval regimes;

– motif enrichment among top-retrieved loci using matched GC/accessibility/length backgrounds;

– overlap with external ChIP-seq peaks for the same or related TFs when available;

– enrichment near genes whose expression changes after TF overexpression;

– stability of retrieved loci across nearby cells or clusters.

Placeholder. Specify motif enrichment tool, motif database, background construction, top-k thresholds, multiple-testing correction, and whether the support definition is same-TF, TF-family, same-TF ReMap, TF-family ReMap, or accessibility-supported motif evidence.

5.6 Variant validation design

Variant experiments are used as sequence-sensitivity diagnostics. The primary comparison is between high-confidence pathogenic or likely pathogenic missense substitutions and matched random single-residue substitutions in the same TF isoform. Matching must control ESM-C distance from wild type, broad protein region, domain class where available, and mutant-residue class where possible. ESM-C distance is an embedding-space quantity, not mutation edit size.

Benign or tolerated variants are not treated as a single negative label. We separate clinical benign variants, population-tolerated variants, functional-assay control variants, ortholog-supported synthetic controls, and legacy conservative fallback controls. These controls are used as secondary analyses only after exact isoform mapping, hard exclusions, and balance diagnostics by TF family, domain class, DNA-binding-domain status, conservation, protein length, recovered cell count, and ESM-C-distance bin.

Benign/control variant construction. The May 17 benign/control table is constructed as an evidence-tiered control set for the local HopTF isoforms, not as a claim that every row is molecularly neutral for TF function. We start from all 3,264 non-control TF isoforms in the local TF Atlas perturbation metadata and create two frozen natural-evidence artifacts: an uncapped table containing all currently mapped ClinVar, gnomAD, and MaveDB rows [Landrum et al., 2025, Karczewski et al., 2020, Esposito et al., 2019], and a capped table containing up to 10 natural benign/control candidates per isoform. Natural human evidence is always used before any synthetic control rows.

Natural variants are accepted only after exact protein-coordinate checking against the local HopTF isoform sequence. The reference amino acid at the one-based variant position must match the local wild-type sequence, and the mutant protein sequence is regenerated from the local sequence rather than copied from the source table. We exclude rows with ambiguous coordinate mapping, non-missense changes, conflicting or uncertain clinical labels, pathogenic or likely pathogenic annotations in the control set, and source records without the information needed to reproduce the protein substitution. Each retained row stores both formal HGVS protein notation, such as p.Arg339Gln, and one-letter shorthand notation, such as R339Q; all HGVS strings in the frozen tables were parsed and sequence-checked.

Table 2: Benign/control evidence tiers used for variant-control construction. Selected rows are the rows used in the regenerated 10-per-isoform capped table. For natural-evidence tiers, uncapped rows are all available exact-mapped rows before the per-isoform cap.

Tier	Inclusion rule	Uncapped rows	Selected rows
Tier 1 clinical benign/control	ClinVar benign, likely benign, or benign/likely benign; no conflicts; exact local isoform match	2,064	1,817
Tier 2 population tolerated, strong	gnomAD v4.1 exome missense row; no disease/conflict annotation; exact local isoform match; group-maximum AF at least 1%	2,734	2,065
Tier 2 population tolerated, moderate	gnomAD v4.1 exome missense row passing the staged population-tolerance filters but below the strong threshold	4,789	1,940
Tier 3 functional-assay control	MaveDB high-score assay row; exact local isoform match; unambiguous high-score-as-more-functional convention; no local ClinVar deleterious match	132	67
Tier 4 ortholog-supported synthetic control	Alternate residue is observed at the aligned position in an Ensembl Compara ortholog; the substitution passes conservative chemistry and local sensitive-position filters; Pfam domain architecture is matched between human and ortholog proteins	–	18,600
Tier 5 legacy conservative fallback control	Deterministic conservative substitution used only when Tiers 1–4 do not provide enough rows for the fixed 10-per-isoform file	–	8,151

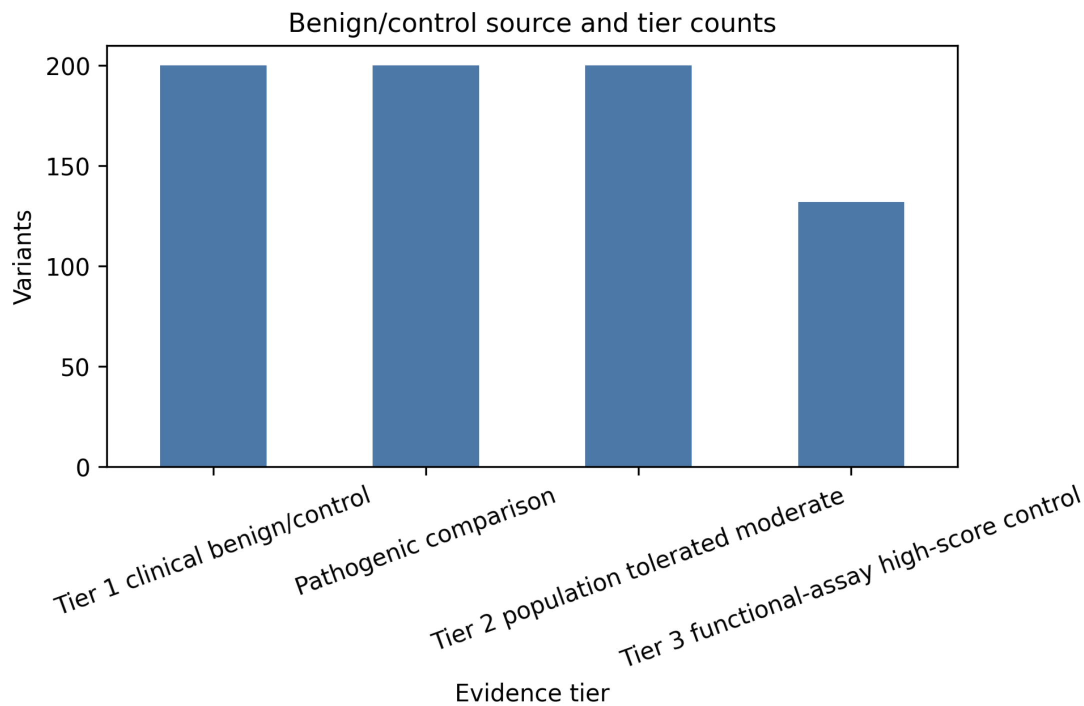

Figure 2: Composition of the evidence-tiered benign/control variant panel. Natural clinical, population, and functional-assay evidence is prioritized first; ortholog-supported tolerated substitutions fill many remaining fixed-size slots; legacy conservative fallback controls are retained only where needed for exactly 10 variants per TF isoform.

The natural-evidence table contains 9,719 rows across 717 isoforms; 435 isoforms have at least 10 natural rows. Because most TF isoforms currently have fewer than 10 natural benign/control rows, we stage a separate tier of ortholog-supported synthetic controls rather than filling the table directly with arbitrary conservative substitutions. For this tier, each proposed mutant residue must be observed at the corresponding aligned position in a non-human Ensembl Compara ortholog [Dyer et al., 2025]. The ortholog protein is aligned back to the local HopTF isoform sequence, the local wild-type residue must match the proposed one-based position, and the mutant sequence is regenerated from the local HopTF sequence. The candidate must pass conservative amino-acid chemistry filters and avoid sequence edges, C2H2-like zinc-finger motifs, basic patches, leucine-zipper-like heptads, low-complexity windows, and cysteine, histidine, glycine, proline, or tryptophan positions when possible. In the final frozen run, we additionally scan the human and ortholog proteins with Pfam [Mistry et al., 2021] and require the same Pfam domain-architecture tuple before accepting the ortholog-supported substitution.

The ortholog-supported generator produced 22,551 candidate substitutions across 2,554 isoforms, with 2,039 isoforms reaching 10 ortholog-supported controls. In the regenerated capped table, natural evidence is selected first, then ortholog-supported tier 4 rows fill the next slots, and tier 5 legacy conservative fallback rows are used only for remaining slots needed to preserve exactly 10 variants for every isoform. The capped table contains 32,640 rows: 5,889 natural-evidence rows, 18,600 ortholog-supported tier 4 rows, and 8,151 tier 5 fallback rows. A total of 2,259 isoforms need no tier 5 fallback.

Tier 4 rows are reported as *ortholog-supported tolerated substitutions* or *evolutionarily supported synthetic controls*. They are not clinical benign variants and are not treated as direct evidence of molecular neutrality. Tier 5 rows are conservative fallback controls. They have no independent clinical, population, functional, or ortholog evidence, and they are included only when Tiers 1–4 do not provide enough rows to maintain the fixed design of 10 control substitutions per TF isoform. Tier 5 rows are therefore software-balancing controls rather than biological evidence; analyses that use them should report them separately and should not interpret them as benign or tolerated variants. The generator records formal HGVS protein notation, one-letter shorthand notation, sequence round-trip validation, and, for tier 4 rows, the supporting ortholog species, ortholog gene/protein identifiers, alignment identity, and alignment coverage. The regenerated local artifacts are stored under `data/processed/variant_controls/tf_isoform_benign_controls_10_per_isoform_ortholog_tier4_20260517/`.

ESM-C distance from wild type is treated as an annotation and matching variable, not as a source of benign labels. Natural controls with large ESM-C shifts are retained only when their independent clinical, population, or functional evidence is strong enough, and they are reported as a stress-test subset rather than silently merged into the core control set. The regenerated capped artifacts are mirrored to `hf://datasets/pvd232/HopTF/processed/variant_controls/tf_isoform_benign_controls_10_per_isoform_ortholog_tier4_20260517`.

5.7 Planned multiple-instance variant training

The current model is trained on one canonical sequence representation per TF isoform, then variants are used only as post hoc sequence perturbations. A more appropriate training formulation is to treat tolerated or benign variants as multiple sequence instances of the same TF isoform. In this planned experiment, each TF isoform would have a bag of sequence instances containing the reference isoform sequence and high-confidence tolerated variants. These instances share the same observed TF Atlas perturbed-cell response, so the model should learn retrieval and response predictions that are stable across tolerated sequence variation. Pathogenic or damaging variants would then be evaluated as held-out sequence instances that may move retrieval or predicted perturbed-cell state activity away from the isoform-level tolerated bag.

This experiment directly matches the weakly supervised multiple-instance structure of HopTF. It asks whether the Hopfield query can be trained to ignore benign protein-sequence variation while remaining sensitive to variants that plausibly alter TF function. It is not run yet, and no result in this draft should be interpreted as evidence for benign-variant invariance until this experiment is designed and executed.

Peter's planned MIL experiment should use the regenerated control table at `hf://datasets/pvd232/HopTF/processed/variant_controls/tf_isoform_benign_controls_10_per_isoform_ortholog_tier4_20260517`. Each variant sequence must be encoded with ESM-C so that the model sees the same type of protein-query input for the reference isoform, tolerated/control instances, and pathogenic held-out variants. The central comparison is:

1. **Reference-only training:** train or evaluate using only the canonical TF isoform sequence.
2. **MIL tolerated-instance training:** train with the canonical sequence plus the Tier 1–4 tolerated/control instances as multiple instances of the same TF isoform response. Tier 5 fallback controls should be reported separately or excluded from the primary MIL claim because they lack independent tolerance evidence.
3. **Pathogenic held-out evaluation:** test whether pathogenic or likely pathogenic variants still produce larger absolute predicted response changes and retrieval changes than tolerated instances.

Figure placeholder: Planned MIL variant-training result

Figure to insert after Peter's run. Suggested panels: left, schematic showing one TF isoform as a bag of reference and tolerated/control protein sequences sharing the same TF Atlas response label; middle, held-out perturbed-cell response prediction before versus after MIL training, where MIL should improve or stabilize prediction relative to reference-only training; right, absolute pathogenic variant effect compared with tolerated/control instances after MIL training, showing that the model remains sensitive to damaging variants rather than becoming invariant to all sequence changes. Caption must state which evidence tiers are used for training and whether Tier 5 fallback rows are excluded from the primary claim.

Figure 3: Planned MIL variant-training result. Figure to insert after Peter's run. Suggested panels: left, schematic showing one TF isoform as a bag of reference and tolerated/control protein sequences sharing the same TF Atlas response label; middle, held-out perturbed-cell response prediction before versus after MIL training, where MIL should improve or stabilize prediction relative to reference-only training; right, absolute pathogenic variant effect compared with tolerated/control instances after MIL training, showing that the model remains sensitive to damaging variants rather than becoming invariant to all sequence changes. Caption must state which evidence tiers are used for training and whether Tier 5 fallback rows are excluded from the primary claim.

6 Results

6.1 Predicting the state of the perturbed cell

We first evaluated prediction of the perturbed-cell state in a conservative PCA-latent benchmark using grouped held-out-gene splits. The purpose of this experiment was not to claim final transport performance, but to ask whether a TF sequence representation still helps to predict the state of the perturbed cell after the model already sees protein length, ORF length, and recovered cell count. The sequence-derived query was produced by projecting the frozen protein embedding into the AlphaGenome key space and retrieving against the same AlphaGenome-derived memory used by the rest of the model.

Table 3: Experiment 1 prediction of the perturbed-cell state on grouped held-out-gene splits. Lower MSE is better.

Feature set	MSE	Median row MSE	Mean cosine similarity
Protein length + ORF length + recovered cell count	6.5996	1.7827	0.3970
ESM-C projected into AlphaGenome key space only	6.8806	1.9395	0.3395
Raw ESM-C embedding	7.3078	2.7258	0.3024
ESM-DBP projected into AlphaGenome key space only	7.9615	3.0910	0.2711
Raw ESM-DBP embedding	8.4732	3.5707	0.2537
Shuffled ESM-C features	8.5951	3.1644	0.0491
Shuffled ESM-DBP features	9.0622	3.3394	0.0571
Training-set mean perturbed-cell state	37.6192	40.6827	-0.4740

The model using only protein length, ORF length, and recovered cell count was difficult to beat. Raw ESM-C and raw ESM-DBP embeddings performed worse than those three quantities alone. The ESM-C-derived representation after projection into AlphaGenome key space was stronger than raw ESM-C, but by itself it still did not beat the model using only protein length, ORF length, and recovered cell count. This motivates the stricter evaluation below, where sequence features are added after the model already sees those three quantities.

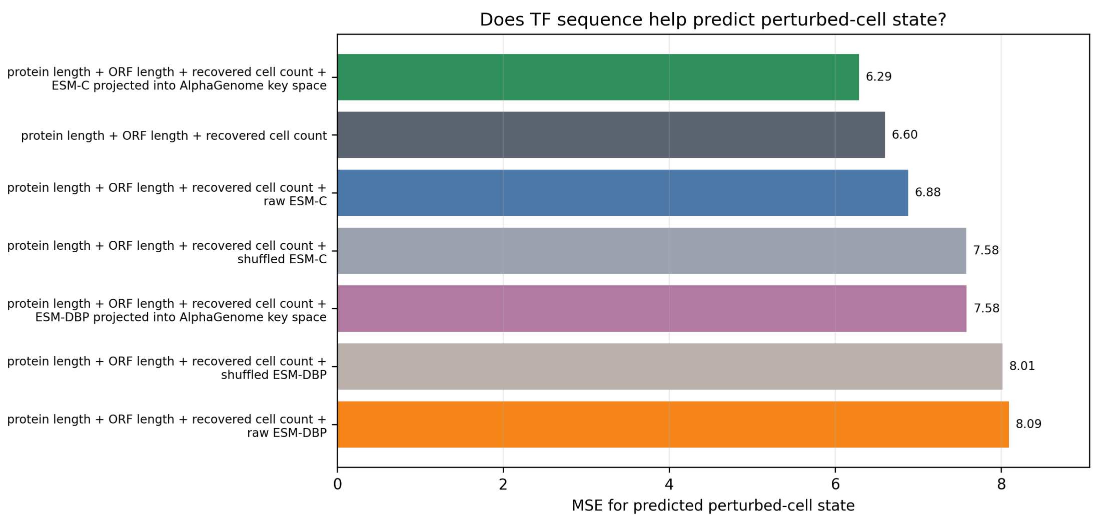

Figure 4: MSE for predicting the perturbed-cell state. The ESM-C-derived representation after projection into AlphaGenome key space modestly improves over using only protein length, ORF length, and recovered cell count. ESM-DBP does not improve this benchmark.

6.2 Ablation studies

Experiment 1B compared the generic ESM-C protein encoder with the DBP-specialized ESM-DBP encoder under the same splits and the same protein-length, ORF-length, and recovered-cell-count inputs. The ESM-DBP embedding generator used the downloaded ESM-DBP checkpoint from the shared cluster path and produced aligned isoform embeddings before fitting the projection into AlphaGenome key space. The ESM-DBP projection required a higher inverse-temperature setting than the default: with $\beta = 30$, 2500 steps, and 64 overfit examples, it reached 100% top-1 retrieval accuracy in the projection smoke test.

Table 4: Overall encoder ablation. Percent change is relative to using only protein length, ORF length, and recovered cell count. Negative values indicate lower MSE.

Feature set	MSE	MSE change vs three-input model	Interpretation
Protein length + ORF length + recovered cell count	6.5996	0.0%	Reference for this comparison
Protein length + ORF length + recovered cell count + ESM-C projected into AlphaGenome key space	6.2879	-4.7%	Lowest MSE in this run
Protein length + ORF length + recovered cell count + raw ESM-C	6.8774	+4.2%	Raw sequence embedding does not help after those three quantities
Raw ESM-C only	7.3078	+10.7%	Weaker than using only protein length, ORF length, and recovered cell count
Protein length + ORF length + recovered cell count + ESM-DBP projected into AlphaGenome key space	7.5832	+14.9%	DBP-specialized representation underperforms ESM-C
Protein length + ORF length + recovered cell count + raw ESM-DBP	8.0926	+22.6%	DBP-specialized raw embedding underperforms ESM-C
Protein length + ORF length + recovered cell count + shuffled ESM-C	7.5761	+14.8%	Negative control worsens as expected
Protein length + ORF length + recovered cell count + shuffled ESM-DBP	8.0145	+21.4%	Negative control worsens as expected

The useful signal in this run appears only after taking the ESM-C embedding and projecting it into the AlphaGenome key space. The raw protein embeddings do not improve prediction of the perturbed-cell state, and ESM-DBP does not replace ESM-C under this comparison. This argues against treating generic protein embeddings as sufficient and supports keeping the retrieval step as the main object of analysis, while also showing that the gains are modest.

The May 17 SCARF-space encoder diagnostic reaches the same encoder conclusion even though the main sequence-feature benchmark is negative. Across all held-out genes and matched settings, positive ESM-DBP-minus-ESM-C MSE differences indicate that ESM-C has lower error than ESM-DBP. For example, in the all-held-out-gene setting, the median ESM-DBP-minus-ESM-C MSE was 4.10×10^{-9} after projection into AlphaGenome key space and 3.47×10^{-9} for raw sequence embeddings. This belongs in the appendix as a reviewer-facing diagnostic rather than a main biological result.

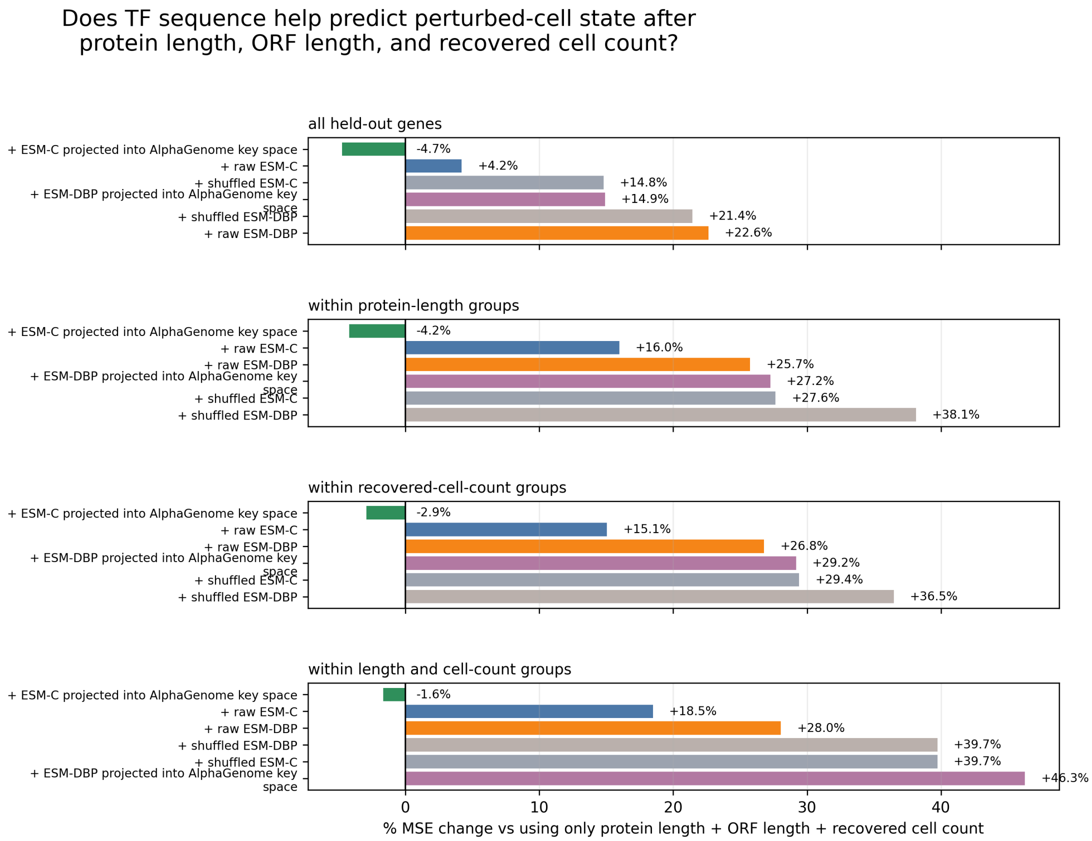

Figure 5: MSE change after adding sequence features to a model that already uses protein length, ORF length, and recovered cell count. ESM-C projected into AlphaGenome key space modestly improves in protein-length groups, recovered-cell-count groups, and joint length-plus-cell-count groups; ESM-DBP is worse in the same comparisons.

6.3 SCARF response representation and controlled sequence benchmark

The May 17 rerun uses SCARF as the primary response representation. For each perturbation, we define the observed response vector as the mean SCARF latent of TF-perturbed cells minus the mean SCARF latent of control cells. The source h5ad contains 671,453 cells, 3,266 perturbation labels, 2,000 control cells, and 512-dimensional SCARF latents. This response representation is not interchangeable with the previous PCA-derived score: SCARF response magnitude is only weakly correlated with recovered cell count ($\rho = -0.098$), protein length ($\rho = -0.075$), and the older PCA response magnitude ($\rho = 0.100$).

The controlled sequence-feature benchmark changes the predictive story. In SCARF space, the model using only protein length, ORF length, and recovered cell count had the lowest held-out-gene MSE among the tested feature sets. Adding the Hopfield query or ESM-C features worsened MSE, and the same pattern held inside protein-length, recovered-cell-count, and joint length-plus-cell-count matched contexts.

Table 5: SCARF-space controlled sequence benchmark. Lower MSE is better. Percent change is relative to the model using protein length, ORF length, and recovered cell count.

Feature set	MSE	MSE change vs length/ORF/cell-count model	Mean cosine similarity
Protein length + ORF length + recovered cell count	8.18×10^{-8}	0.0%	0.8402
Protein length + ORF length + recovered cell count + Hopfield query	8.42×10^{-8}	+2.9%	0.8317
Protein length + ORF length + recovered cell count + ESM-C	8.72×10^{-8}	+6.5%	0.8291
Hopfield query only	8.74×10^{-8}	+6.8%	0.8326
ESM-C only	9.11×10^{-8}	+11.3%	0.8299
Shuffled ESM-C after length/ORF/cell count	9.42×10^{-8}	+15.2%	0.8290
Training-set mean response	3.65×10^{-7}	+346.0%	0.0000

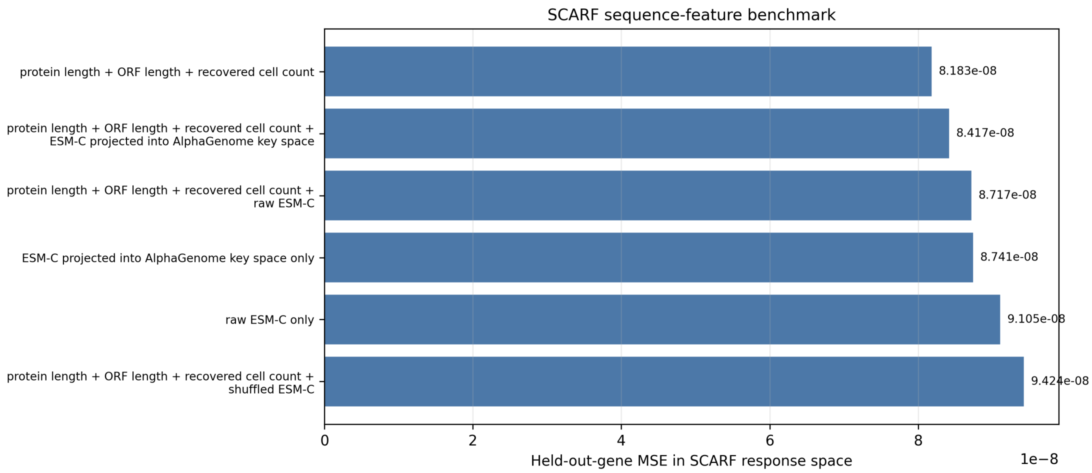

Figure 6: SCARF-space controlled sequence-feature benchmark. Protein length, ORF length, and recovered cell count are difficult to beat; adding the Hopfield query or ESM-C features does not improve held-out-gene prediction in this response representation.

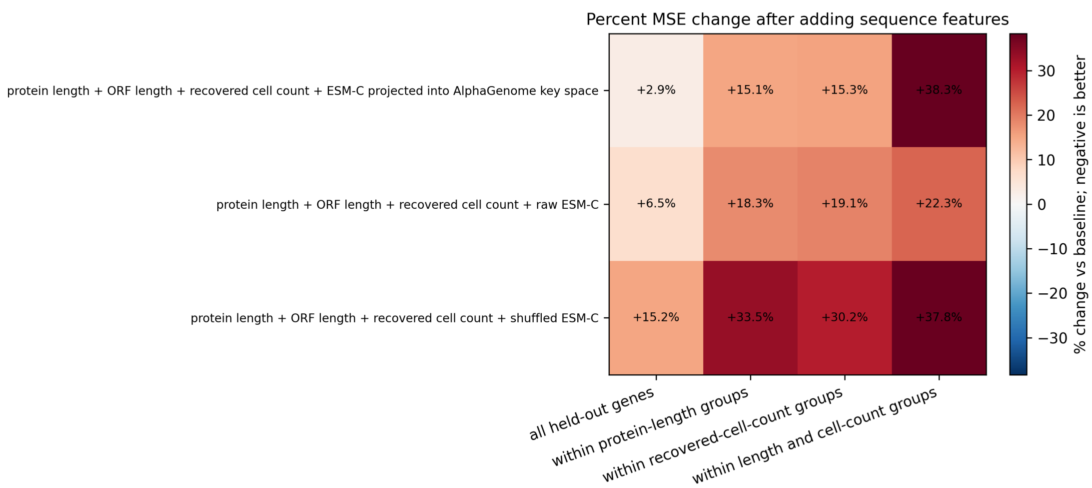

Figure 7: Matched-context SCARF-space sequence benchmark. The heatmap diagnoses whether sequence features help after stratifying by protein length, recovered cell count, or both; in this run, the sequence-feature additions do not improve over the measured covariates.

This is a negative result for the strongest version of the predictive claim. The earlier PCA-space benchmark remains useful as a sensitivity analysis, but the primary SCARF-space benchmark does not support claiming that the current sequence or Hopfield-query features improve held-out TF response prediction after measured non-sequence variables are included. The paper should therefore shift emphasis from broad predictive performance to controlled sequence-sensitivity and retrieval diagnostics.

Interpretation placeholder. Do not describe SCARF as automatically more biological than PCA. The relevant question is whether HopTF's sequence-conditioned conclusions persist after changing the cell-state representation while still controlling for protein length, ORF length, and recovered cell count. Here the predictive benchmark does not persist, so the PCA result must be framed as a sensitivity result rather than the main response-prediction claim.

6.4 Missense variants perturb predicted TF response

We next used TF missense variants as a sequence sensitivity diagnostic. This analysis does not ask whether HopTF classifies clinical variants. Instead, it asks whether changing a TF amino acid sequence changes the predicted perturbation response more than matched control substitutions. For each variant, we compared the predicted response for the wild type isoform with the predicted response for the mutant isoform in SCARF latent space. We report both signed changes, which test whether a variant weakens the predicted response, and absolute changes, which test whether a variant changes the predicted response in either direction.

Pathogenic variants did not consistently reduce the predicted response relative to wild type. This is not surprising: damaging TF variants may weaken, strengthen, or redirect a transcriptional response depending on the affected domain and cellular context. The stronger result is therefore the absolute disruption test. Pathogenic variants produced larger absolute predicted response changes than matched random substitutions, with 189 matched sets, median paired difference 0.0111, and one-sided Wilcoxon $p = 0.0084$. Pathogenic variants also produced larger absolute predicted response changes than matched benign/control variants, with 158 matched sets, median paired difference 0.0346, and one-sided Wilcoxon $p = 5.46 \times 10^{-6}$.

Table 6: SCARF-space matched variant validation. Positive paired differences mean pathogenic variants changed predicted perturbed-cell state activity more than the matched control group.

Comparison	Metric	Matched sets	Median paired difference	Fraction pathogenic greater	Wilcoxon p
Pathogenic vs. matched random substitutions	Signed weakening	189	0.0138	0.5503	0.9735
Pathogenic vs. matched random substitutions	Absolute disruption	189	0.0111	0.5397	0.0084
Pathogenic vs. evidence-tiered benign/control	Signed weakening	158	0.0451	0.5823	0.9996
Pathogenic vs. evidence-tiered benign/control	Absolute disruption	158	0.0346	0.5949	5.46×10^{-6}

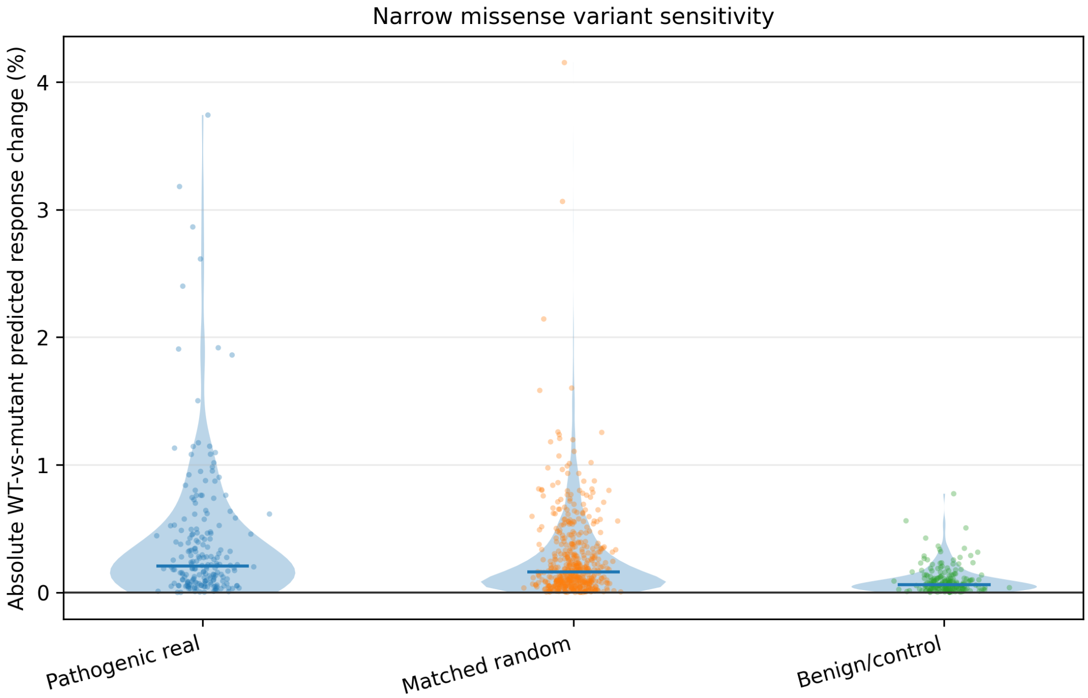

Figure 8: Absolute predicted perturbed-cell state activity change for pathogenic variants versus matched controls. This is the primary variant-sensitivity framing because damaging TF variants may weaken, strengthen, or redirect predicted response rather than only reducing response magnitude.

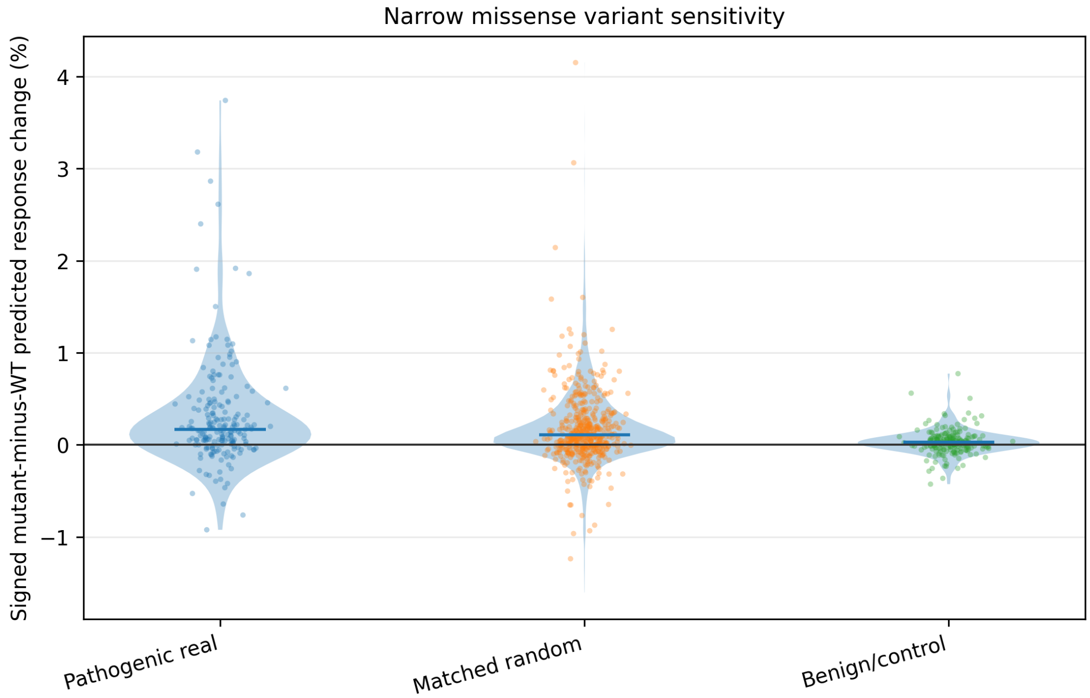

Figure 9: Signed predicted perturbed-cell state activity change for pathogenic variants versus matched controls. This diagnostic shows why the paper does not claim a consistent signed weakening effect: pathogenic variants are not expected to move responses in one direction only.

Pathogenic and control distributions still overlap substantially: 0.5397 of pathogenic variants exceeded their matched random controls, and 0.5949 exceeded their matched benign/control variants. These results show that TF missense substitutions can move HopTF's predicted response away from the wild type prediction, but they do not show that individual pathogenic variants are consistently distinguishable from matched controls. Matched random substitutions provide a useful and straightforward reference because they ask whether a pathogenic single residue substitution moves the predicted response more than another single residue change in the same isoform. The matched benign/control comparison points in the same direction, but it should be treated as provisional because this run used an earlier, less complete benign/control panel than the regenerated panel in Section 5.6. This motivates the regenerated benign/control design in Section 5.6, which builds up to 10 control substitutions for each TF isoform by prioritizing natural clinical, population, and functional evidence, then filling remaining slots with ortholog-supported tolerated substitutions and, only when necessary, conservative fallback controls.

Interpretation placeholder. ESM-C distance from wild type is an embedding-space matching variable, not mutation edit size. A single amino-acid substitution can have a large ESM-C distance, and high ESM-C distance should be diagnosed rather than silently equated with a large mutation.

6.5 Variant effects move the retrieval distribution

We next asked whether variants that alter predicted response also change the model's retrieval pattern over AlphaGenome windows. This is a diagnostic of HopTF's intermediate representation:

if sequence changes affect predicted cell state only after changing retrieval, then the model exposes an auditable link between TF protein sequence and the genomic context used for prediction. This analysis should therefore be interpreted as evidence that missense substitutions can propagate through HopTF's retrieval layer, changing which genomic contexts are prioritized before prediction. It does not show that the TF actually binds, stops binding, or changes binding strength at any specific genomic location.

Table 7: Matched retrieval-change diagnostic. Positive paired differences mean pathogenic variants moved the retrieval distribution more than the matched control group.

Comparison	**Matched sets**	**Median paired difference**	**One-sided Wilcoxon p**
Pathogenic vs. matched ClinVar benign	158	4.34×10^{-5}	0.2478
Pathogenic vs. matched random substitutions	189	1.08×10^{-4}	2.62×10^{-4}

Pathogenic variants moved the retrieval distribution more than matched random substitutions, with 189 matched sets, median paired difference 1.08×10^{-4}, and one-sided Wilcoxon $p = 2.62 \times 10^{-4}$. The comparison against matched ClinVar benign variants was weaker, with 158 matched sets, median paired difference 4.34×10^{-5}, and one-sided Wilcoxon $p = 0.2478$. This mirrors the response analysis: matched random substitutions provide a useful reference for same-isoform sequence perturbation, while benign or tolerated controls require the stronger control-panel design in Section 5.6.

Figure 10 shows selected examples as retrieval weights over AlphaGenome windows arranged by chromosome for wild type, pathogenic mutant, and matched benign/control sequences. The difference panel shows where retrieval weight increases or decreases after the pathogenic sequence change. These panels make the intermediate representation inspectable, but they do not identify measured binding sites or causal regulatory targets.

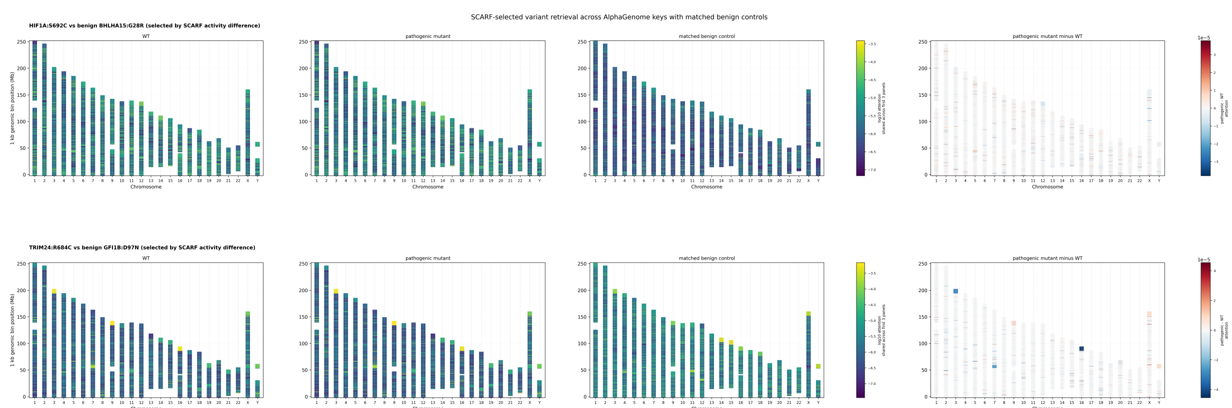

Figure 10: Variant-specific retrieval movement across AlphaGenome windows. Each chromosome is split into the window bins used by HopTF. The panels show wild type, pathogenic mutant, matched benign/control sequence, and pathogenic minus wild type change for selected examples. These are retrieval weights over locus windows, not measured binding sites.

6.6 β controls retrieval regime

A sweep over retrieval temperature tests whether the HopTF retrieval layer behaves like a controllable associative retrieval system. As β increases from 0.1 to 30, retrieval moves from diffuse averaging over many AlphaGenome-key windows toward more focused use of a smaller subset. The median effective number of retrieved loci falls from 5.49×10^4 to 5.21×10^3, while median top-100 mass rises from 0.0019 to 0.2166. At the same time, adjacent β values have median top-100 overlap of 1.0, indicating that β mostly sharpens a stable ranking rather than replacing the highest-ranked loci.

This result supports the intended associative retrieval interpretation: β acts as an inverse retrieval temperature that controls how broadly or selectively HopTF prioritizes regulatory loci. This makes β a hyperparameter with a biologically interpretable role for researchers studying transcription factors. A researcher studying a specific TF can inspect retrieval across lower and higher β values to compare broad regulatory-context averaging with more focused candidate-region hypotheses. Prior knowledge about TF specificity, chromatin context, or expected regulatory breadth can guide which regime is most useful for exploration.

This interpretation remains qualitative. The β sweep does not show that β is a dose-calibrated proxy for TF concentration, that higher-β retrieval is more biologically correct, or that any β value improves perturbed-state prediction. Instead, β should be used as a sensitivity parameter for hypothesis generation: regions or patterns that are stable across β values are stronger candidates for follow-up, while β-specific regions may indicate hypotheses that depend on assuming more diffuse or more selective regulatory prioritization.

Table 8: β controls retrieval concentration. Effective loci and top-100 mass are medians across TF rows.

β	**Median effective loci**	**Median top-100 mass**
0.1	5.49×10^4	0.0019
0.3	5.49×10^4	0.0020
1	5.48×10^4	0.0023
3	5.37×10^4	0.0036
10	4.24×10^4	0.0144
30	5.21×10^3	0.2166

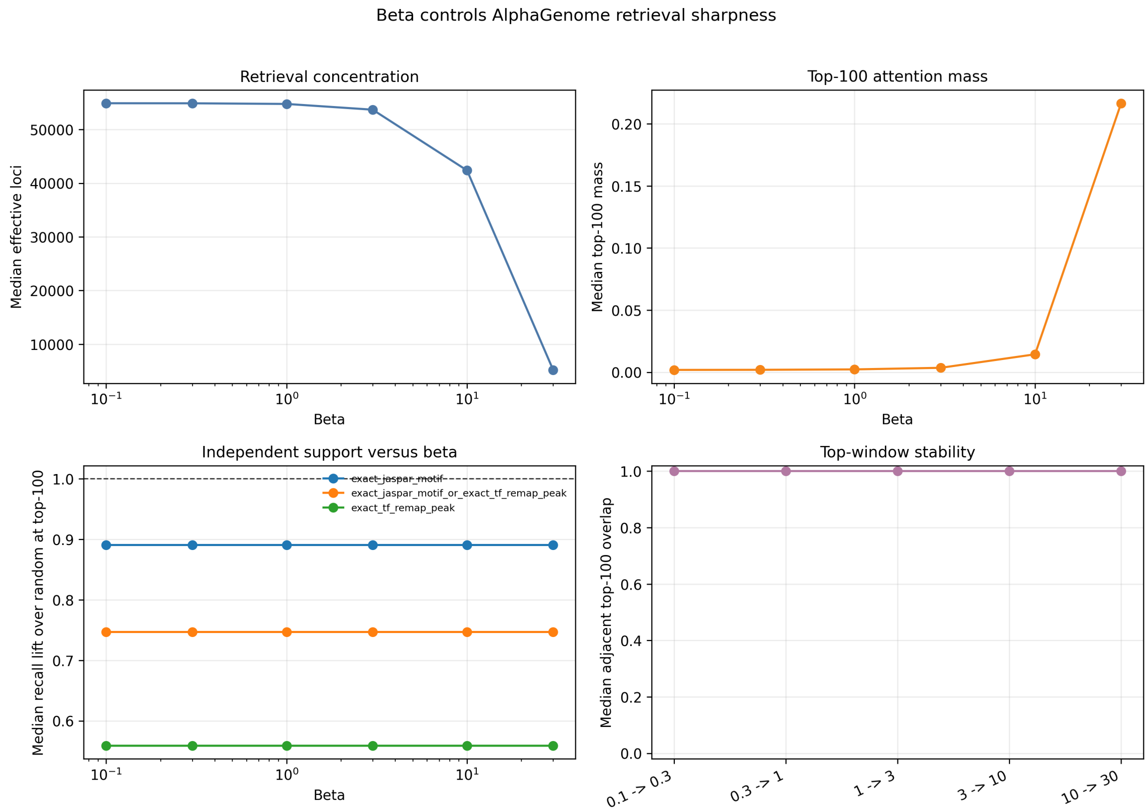

Figure 11: Effect of retrieval inverse temperature β. Increasing β shifts retrieval from diffuse averaging over many AlphaGenome windows toward more focused use of a smaller subset, while adjacent β values preserve the same top-ranked windows. Independent motif and ReMap support remains weak across the sweep. The β result supports controllable associative retrieval behavior and a biologically interpretable sensitivity parameter, not dose-calibrated TF concentration or validated binding-site discovery.

Interpretation placeholder. This result supports controllable associative retrieval behavior, not a dose-calibrated biological concentration claim. Safe phrasing: β is an inverse retrieval temperature that lets researchers explore broader or more focused regulatory-locus prioritization. The current run does not test whether any β value improves SCARF-space perturbed-cell prediction, because the SCARF transport checkpoint was not β-parameterized.

6.7 Retrieval distributions identify candidate compatible regulatory loci

Placeholder. Optional late-arriving retrieval case-study slot. This is the Audrey/main-figure retrieval-annotation slot, not the planned Peter MIL variant-training extension described in Section 5.6. Fill only if selected TFs can be supported with top retrieved AlphaGenome windows, nearby genes, motif/ReMap or accessibility support, and a conservative perturbation-response interpretation. If this result is not ready before submission, remove this subsection and rely on the aggregate variant, retrieval movement, β, and motif/ReMap diagnostics.

Table 9: Placeholder for optional TF-specific retrieval case studies.

TF	**Top retrieved loci / annotations**	**Motif or ChIP evidence**	**Perturbation-response interpretation**
`[TF 1]`	[TODO]	[TODO]	[TODO]
`[TF 2]`	[TODO]	[TODO]	[TODO]
`[TF 3]`	[TODO]	[TODO]	[TODO]

Figure placeholder: Optional retrieval case-study tracks

Plot to insert only if retrieval annotations are ready. A useful panel would show retrieval weights over loci for selected TFs and cell-state clusters, with tracks for accessibility, motif hits, nearby genes, and observed expression changes if available.

Figure 12: Optional retrieval case-study tracks. Plot to insert only if retrieval annotations are ready. A useful panel would show retrieval weights over loci for selected TFs and cell-state clusters, with tracks for accessibility, motif hits, nearby genes, and observed expression changes if available.

Interpretation placeholder. If this subsection is filled, use cautious language: prioritized windows are candidate regulatory contexts. They are not validated binding sites unless supported by independent evidence. If the case-study analysis is not complete, remove this placeholder subsection before final submission rather than leaving TODO text in the PDF.

6.8 Motif and ReMap support remains limited under exact matching

We next asked whether the windows prioritized by HopTF overlap independent motif or binding evidence for the same TF. We used JASPAR as a curated database of TF DNA binding motifs [Rauluseviciute et al., 2024] and ReMap as an external compendium of TF ChIP-seq peaks [Hammal et al., 2022]. For motif and ReMap validation, the unit of evaluation is a 1 kb window centered on the TSS associated with each AlphaGenome key locus, not the exact motif base or binding event. A supported window is therefore a broad regulatory neighborhood containing a JASPAR motif match assigned to the same TF symbol or overlapping a ReMap peak assigned to the same TF. This test can make a retrieved window more plausible, but it cannot show that the TF binds the exact site in TF Atlas cells. Here, top k refers to the k AlphaGenome windows assigned the highest retrieval weights for a given TF.

We use two thresholds because the diagnostics answer different questions. The top 100 retrieved windows represent the most prioritized hypotheses and are useful for asking whether the strongest model nominations are enriched for motif or peak support. The top 1000 retrieved windows provide a broader recall test, asking whether known motif or ReMap supported windows appear anywhere near the upper part of the retrieval ranking despite sparse annotations and large AlphaGenome windows.

Under this exact TF symbol matching criterion, the current validation provides only limited support for motif or binding recovery. In the recall analysis, JASPAR support among the top 1000 retrieved windows had median recall of 1.77% across 1616 TF rows, close to 1.82%, the value expected if supported windows were distributed uniformly across the retrieval ranking. ReMap support among the top 1000 retrieved windows had median recall of 1.13% across 1071 TF rows, also below the value expected under a random ordering of windows, and the union of JASPAR or ReMap support had median recall of 1.47%. Thus, retrieved windows do not recover a larger fraction of exact same-TF motif or binding evidence than expected under this baseline.

We also asked whether the top 100 retrieved windows were more often supported than matched background windows. This result was modest: the median support fraction was 0.060 in retrieved windows and 0.055 in matched background windows, with only 5 TF rows passing FDR ≤ 0.10. This suggests that some individual TF rows may contain useful motif or peak signal, but the overall exact symbol analysis does not support a broad claim that HopTF has learned binding maps. Broad regulatory-annotation overlap is useful for showing regulatory context, but it does not establish that the queried TF binds that locus in TF Atlas cells.

Taken together, these analyses show that HopTF can nominate regulatory hypotheses, but that their biological support remains unresolved at the current resolution. The 1 kb validation windows are much larger than TF motifs, ReMap peaks come from external cell contexts, and exact TF symbol matching can miss aliases, closely related TFs with similar binding preferences, cofactors, and chromatin context. A stronger validation would relax exact symbol matching where biologically justified, incorporate accessibility support, and compare retrieved windows against matched background windows. Until those tests improve, prioritized windows should be described as candidate regulatory contexts, not validated binding sites or causal targets.

Table 10: Motif and ReMap recall remains limited under exact TF symbol matching; stronger validation with accessibility support and matched background windows remains pending.

Support definition	**Top-k**	**TF rows**	**Median recall@k**	**Expected if ranks were random**
Exact same symbol JASPAR motif	1000	1616	0.0177	0.0182
Exact same TF ReMap peak	1000	1071	0.0113	0.0182
Exact JASPAR motif or exact TF ReMap peak	1000	1616	0.0147	0.0182

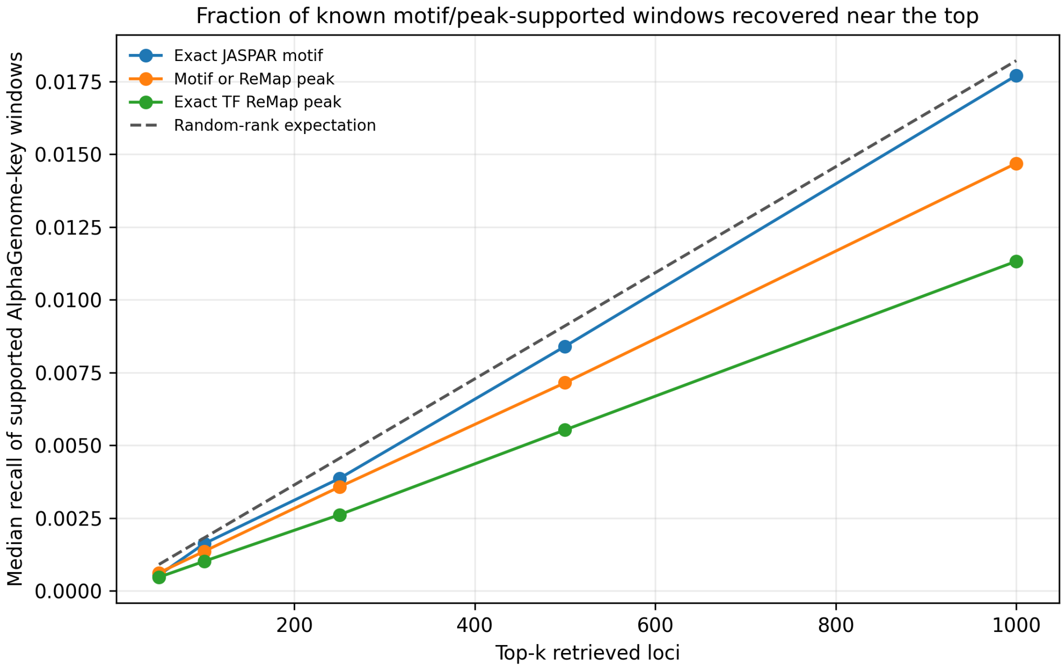

Figure 13: Recall of exact motif and ReMap support among top retrieved windows. Recall remains near or below the value expected if the retrieval ranking were random at the current 1 kb validation-window scale, so this result should be interpreted as a limitation rather than positive binding validation.

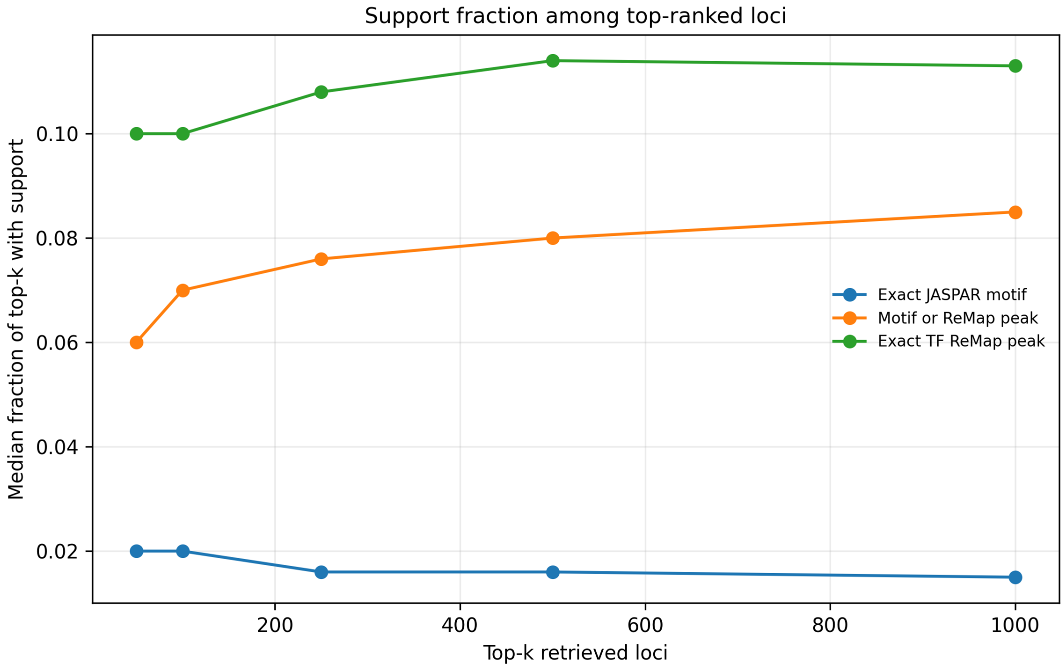

Figure 14: Motif and ReMap support among top retrieved windows. This diagnostic asks what fraction of retrieved windows have exact TF symbol support; it remains modest and motivates validation that relaxes exact symbol matching where biologically justified and incorporates accessibility support.

Interpretation placeholder. Do not force a mechanistic interpretation from the current exact symbol recall results. Possible explanations include the mismatch between short motifs and 1 kb validation windows, sparse or alias dependent TF motif mappings, missing cell-state context, cofactors that do not bind DNA directly, inadequate memory embeddings, or perturbed-cell-state supervision dominated by dataset structure.

6.9 Natural isoform diagnostic

We also tested whether natural same-gene isoform differences with interpretable domain annotations show retrieval changes that track observed response-score differences. The current result should remain diagnostic rather than a main claim. In the conservative domain-aware rerun, 246 same-gene isoform pairs had unavailable annotations and 9 pairs had no annotated domain-presence difference. This coverage is too limited to conclude that natural isoform domain differences do not matter.

Placeholder. After the isoform annotation audit, either replace this paragraph with a domain-aware isoform result using UniProt, InterPro, Pfam, Ensembl/GENCODE, and TF-specific annotations, or move the diagnostic to the appendix and state that feature-mapping coverage was insufficient for a main isoform experiment.

Interpretation placeholder. The negative isoform diagnostic currently says more about annotation coverage and coordinate mapping than about isoform biology. Do not claim that natural isoform differences fail to affect retrieval unless a redesigned annotation audit supports that conclusion.

6.10 PCA sensitivity checks for protein length, ORF length, and recovered cell count

Predicting TF Atlas perturbed-cell states is strongly affected by protein length, ORF length, and recovered cell count. Longer ORFs and proteins are harder to package and infect in lentiviral libraries, reducing recovered cell counts. Separately, strong responders can reduce recovered cells because cells die or leave the assayed state. These mechanisms interact so that protein length, recovered cell count, and response score are correlated in the observed data. We therefore model protein length, ORF length, and recovered cell count directly rather than treating them as incidental metadata. The table below comes from the older PCA-derived response representation and should be read as a sensitivity check, not as the primary SCARF-space result.

Table 11: Checks using protein length, ORF length, and recovered cell count. Lower MSE is better. The matched rows evaluate whether sequence-derived features add signal within groups defined by protein length and recovered cell count.

Control	MSE	Mean cosine similarity	Conclusion
Protein length only	6.6355	0.4114	Strong by itself
Protein length + cell count	6.6020	0.3994	Nearly matches length+cell+ORF
Protein length + ORF length + recovered cell count	6.5996	0.3970	Reference for the sequence-feature comparisons
Cell count only	7.2557	0.3161	Weaker alone than length-based covariates
ESM-C projected into AlphaGenome key space, protein-length groups	6.2714	0.4716	4.2% lower MSE
ESM-C projected into AlphaGenome key space, cell-count groups	6.4074	0.3878	2.9% lower MSE
ESM-C projected into AlphaGenome key space, length+cell groups	6.4603	0.4604	1.6% lower MSE
ESM-DBP projected into AlphaGenome key space, length+cell groups	9.6068	0.3185	Worse

These controls are central to the interpretation. Protein length, recovered cell count, and ORF length explain enough of the perturbed-cell state that raw sequence features alone are not convincing. In the older PCA-derived response space, adding the ESM-C-derived representation after projection into AlphaGenome key space modestly improved MSE. In the primary SCARF-space rerun, that improvement did not persist. The safe interpretation is that the current predictive sequence signal is response-representation dependent, while the stronger current evidence for sequence sensitivity comes from the matched variant analyses.

7 Discussion

7.1 What HopTF adds beyond "slapping on a transformer"

HopTF does not claim novelty from the softmax operation itself. The retrieval layer is mathematically related to attention, and modern Hopfield networks explicitly connect attention to associative memory. The contribution is the biological decomposition: protein sequence provides the query, genomic loci provide the addressable memory, retrieval produces a regulatory-context hypothesis, and a conditional flow model maps that retrieved representation to a predicted cell-state shift. A generic transformer perturbation model may improve expressivity, but it typically does not specify the biological meaning of queries, keys, values, masks, retrieval temperatures, or retrieval diagnostics.

This distinction matters because HopTF exposes a falsifiable intermediate object. The locus activation distribution can be audited against motifs, accessibility, target-gene annotations, ChIP-seq evidence, protein length, ORF length, recovered cell count, and shuffled sequence representations. Predicting the perturbed-cell state alone is not sufficient evidence that the model has learned biologically meaningful regulation. The retrieval distribution is therefore both the mechanism and the liability of the model: it creates interpretability, but it must be validated.

7.2 Associative memory as a modeling lens

The memory in HopTF is the locus-indexed key/value bank, not the transport predictor. The transport predictor is the downstream flow module that uses the retrieved memory state. In associative-memory terms, genomic locus embeddings are stored patterns, the TF isoform embedding is the retrieval cue, and μ_p is the retrieved regulatory-context prototype. This framing connects HopTF to the memory-model side of the course. The model studies how a biological perturbation cue can retrieve information from a large structured memory under weak aggregate supervision.

The model is not a continual-learning algorithm in its current form. However, the explicit memory-bank formulation suggests continual extensions. New cell types, accessibility profiles, perturbation datasets, or genomic annotations could update or augment the regulatory memory while preserving the retrieval-and-transport interface. Robust continual updating would still require mechanisms such as replay, regularization, adapters, or frozen base memories to avoid catastrophic forgetting.

7.3 Biological interpretation and hypothesis generation

HopTF is best interpreted as a hypothesis-generation model. We tolerate false positives more than false negatives if the goal is to nominate candidate compatible loci for follow-up analysis. This changes the evaluation standard. The model does not need every high-weight locus to be causal, but it should enrich for plausible regulatory contexts, avoid obvious confounding by dataset structure, and recover known motifs or target programs when strong prior evidence exists.

A successful mature result would look like this: predictions of perturbed-cell state are competitive with simple comparisons; retrieval is not collapsed or random; top-retrieved windows are enriched for same-TF, TF-family, or accessibility-supported motif and binding evidence; retrieved windows are plausible under accessibility or cluster context; and case studies produce testable regulatory hypotheses. The current result is more mixed. The SCARF-space predictive benchmark is not improved by sequence features after protein length, ORF length, and recovered cell count are included. However, the variant experiments show sequence sensitivity under an absolute-disruption framing, and the β sweep confirms that the retrieval module exposes a controllable concentration parameter. This combination is useful, but it is not enough to claim that the current model has learned a validated regulatory mechanism.

7.4 Limitations

Several limitations should be explicit.

1. **Weak supervision.** The model is trained on aggregate perturbation responses, not locus-level binding or causal regulatory labels. Retrieval weights are therefore not binding probabilities.

2. **Cell-state context.** Fully cell-specific accessibility masking may be unrealistic if accessibility is not observed or accurately inferred. Cluster-specific biases may be more stable but less granular.

3. **Pretrained embedding dependence.** If AlphaGenome or protein embeddings fail to encode the relevant TF-locus compatibility information, the retrieval module may learn shortcuts.

4. **Dataset structure.** Perturbation cell counts, ORF length, TF family identity, baseline expression, and barcode effects may explain perturbed-cell-state metrics unless explicitly controlled.

5. **Variant-label ambiguity.** Clinical benignity, population tolerance, and assay-level molecular neutrality are different evidence types. Benign variants should therefore be curated and reported by evidence tier rather than pooled into one negative label.

6. **Response-representation dependence.** The older PCA-derived benchmark suggested modest sequence-feature gains, but the primary SCARF-space benchmark did not. This limits any claim that the present sequence representation improves general perturbed-state prediction.

7. **Resolution mismatch.** TF motifs are often only several to tens of base pairs, while the current AlphaGenome-key windows are much larger. Window-level retrieval can therefore miss exact motif recall even when it nominates broad regulatory neighborhoods.

8. **Computational scale.** Genome-wide retrieval over millions of loci may require approximate retrieval, candidate pruning, hierarchical memory, or region-level prefiltering.

9. **Causal ambiguity.** Even motif-enriched high-retrieval loci may be passengers or correlated contexts rather than causal mediators of the transcriptional response.

7.5 Future directions

Natural extensions include adapter-based task adaptation of AlphaGenome-derived memory slots, hierarchical retrieval from chromosomes to loci, explicit enhancer-gene linking in value vectors, auxiliary motif or ChIP supervision, dose-conditioned retrieval temperature calibration, and continual updates to the memory bank as new cell types and perturbation datasets become available. A stronger biological model would also distinguish direct TF binding, cofactor-mediated regulation, chromatin remodeling, and downstream secondary response programs.

The most important planned training extension is the multiple-instance variant experiment described above. Instead of treating benign or tolerated variants only as post hoc controls, the model would train on multiple tolerated sequence instances per TF isoform that share the same observed TF Atlas response. This should improve the response model by using more sequence instances per TF while explicitly teaching the query and retrieval system to be stable to tolerated sequence variation. The key success criterion is not only lower held-out response error: pathogenic variants should still produce larger absolute predicted response or retrieval changes than tolerated instances after MIL training. That combination would support the intended distinction between tolerance-aware invariance and loss of sequence sensitivity.

8 Conclusion

HopTF formulates TF perturbation prediction as sequence-conditioned associative retrieval over a genome-indexed regulatory memory. A TF isoform sequence embedding queries AlphaGenome-derived locus embeddings, optionally modulated by cell-state context, and the resulting activation distribution over loci is pooled into a perturbation-conditioning representation μ_p. This representation conditions a flow model that predicts the transport from control to TF-perturbed cell states. The model is end-to-end differentiable through retrieval and response prediction, allowing aggregate perturbation supervision to shape the addressing function even without locus-level labels. The central claim is therefore not that attention alone is novel, but that biologically specified associative retrieval creates an interpretable, falsifiable intermediate between TF sequence and cell-state response.

The current experiments make this claim more precise. SCARF-space perturbed-state prediction is not improved by adding sequence features after measured covariates, so the draft should not overstate predictive performance. The strongest positive evidence is that pathogenic missense variants cause larger absolute predicted response changes than matched controls, and that the Hopfield inverse temperature controls retrieval concentration as expected. The main remaining limitation is biological validation: exact-symbol motif and ReMap support at the current AlphaGenome-window scale is weak, so high-retrieval windows remain hypotheses rather than validated binding sites.

A Appendix: Suggested Figures

1. **Figure 1: Model overview.** TF sequence query, AlphaGenome locus memory, cell-state bias/mask, retrieval weights, μ_p, conditional flow field.

2. **Control panel construction.** Evidence-tier counts for the benign/control variant panel, with natural evidence, ortholog-supported tolerated substitutions, and legacy fallback controls separated.

3. **Controlled perturbed-cell state prediction.** Primary SCARF-space benchmark after controlling protein length, ORF length, and recovered cell count, with PCA-derived results clearly labeled as sensitivity analysis.

4. **Variant sequence sensitivity.** Absolute disruption for pathogenic variants against matched random substitutions as the main panel; signed weakening and evidence-tiered benign/control variants as secondary panels.

5. **Variant retrieval movement.** Chromosome-binned AlphaGenome-window attention for WT, pathogenic mutant, and matched benign/control sequences, plus pathogenic-minus-WT signed change.

6. **MIL variant training.** Schematic and result panel showing that training on multiple tolerated/control sequence instances per TF improves or stabilizes response prediction while preserving larger pathogenic variant effects.

7. β **sweep.** Retrieval concentration, top-k mass, and retrieval stability across β. Include perturbed-cell-state performance only if a β-parameterized SCARF transport checkpoint is run.

8. **Motif and binding support.** Precision@k, recall@k, rank-percentile distributions, and matched-background enrichment for exact-symbol, TF-family, ReMap, and accessibility-supported windows when the family/alias maps are reliable.

9. **Appendix figure: ESM-C vs ESM-DBP.** SCARF-space encoder diagnostic showing that ESM-DBP does not outperform ESM-C in this setup.

10. **Appendix figure: Length, cell-count, and shuffled-sequence checks.** Performance of length/cell-count inputs and shuffled-sequence comparisons, labeled by response representation.

B Appendix: Reporting Checklist

– State whether AlphaGenome, ESM-C, and SCARF are frozen, adapter-tuned, or fully fine-tuned.

– State whether values are tied to keys.

– State whether retrieval is unmasked, cluster-masked, or cell-specific.

– Report primary perturbed-cell-state metric and confidence intervals.

– Report retrieval entropy, top-k mass, and β sensitivity.

– Include at least one comparison that uses protein length, ORF length, and recovered cell count.

– Report whether conclusions persist in SCARF response space or remain specific to the PCA-derived response representation.

– Define variant labels by evidence tier; do not pool clinical benign, population-tolerated, functional-neutral, and synthetic controls without showing tier-specific behavior.

– Define motif/binding-supported windows at the AlphaGenome-window level and state the genome build, support database, background construction, and multiple-testing correction.

– Avoid interpreting retrieval weights as binding sites without external validation.

References

Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. *Nature Methods*, 22 (8):1657–1661, 2025. doi: 10.1038/s41592-025-02772-6.

Žiga Avsec, Natasha Latysheva, Jun Cheng, Guido Novati, Kyle R. Taylor, Tom Ward, Clare Bycroft, Lauren Nicolaisen, Eirini Arvaniti, Joshua Pan, Raina Thomas, Vincent Dutordoir, Matteo Perino, Soham De, Alexander Karollus, Adam Gayoso, Toby Sargeant, Anne Mottram, Lai Hong Wong, Pavol Drotár, Adam Kosiorek, Andrew Senior, Richard Tanburn, Taylor Applebaum, Souradeep Basu, Demis Hassabis, and Pushmeet Kohli. Advancing regulatory variant effect prediction with AlphaGenome. *Nature*, 649(8099):1206–1218, 2026. doi: 10.1038/s41586-025-10014-0.

Shreya Basnet and Jianlin Cheng. Integrating protein and DNA embeddings for improving genome-wide transcription factor binding site prediction. *NAR Genomics and Bioinformatics*, 8(2): lqag047, 2026. doi: 10.1093/nargab/lqag047.

Carmen Bravo González-Blas, Seppe De Winter, Gert Hulselmans, Nikolai Hecker, Irina Matetovici, Valerie Christiaens, Suresh Poovathingal, Jasper Wouters, Sara Aibar, and Stein Aerts. SCENIC+: single-cell multiomic inference of enhancers and gene regulatory networks. *Nature Methods*, 20(9):1355–1367, 2023. doi: 10.1038/s41592-023-01938-4.

Charlotte Bunne, Stefan G. Stark, Gabriele Gut, Jacobo Sarabia del Castillo, Mitch Levesque, Kjong-Van Lehmann, Lucas Pelkmans, Andreas Krause, and Gunnar Rätsch. Learning single-cell perturbation responses using neural optimal transport. *Nature Methods*, 20(11):1759–1768, 2023. doi: 10.1038/s41592-023-01969-x.

Haotian Cui, Chloe Wang, Hassaan Maan, Kuan Pang, Fengning Luo, Nan Duan, and Bo Wang. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, 21(8):1470–1480, 2024. doi: 10.1038/s41592-024-02201-0.

Sarah C. Dyer, Olanrewaju Austine-Orimoloye, Andrey G. Azov, Matthieu Barba, If Barnes, Vianey Paola Barrera-Enriquez, Arne Becker, Ruth Bennett, Martin Beracochea, Andrew Berry, et al. Ensembl 2025. *Nucleic Acids Research*, 53(D1):D948–D957, 2025. doi: 10.1093/nar/gkae1071.

ESM Team. ESM Cambrian: revealing the mysteries of proteins with unsupervised learning. EvolutionaryScale website, December 2024. URL https://www.evolutionaryscale.ai/blog/esm-cambrian. Accessed 18 May 2026.

Daniel Esposito, Jochen Weile, Jay Shendure, Lea M. Starita, Anthony T. Papenfuss, Frederick P. Roth, Douglas M. Fowler, and Alan F. Rubin. MaveDB: an open-source platform to distribute and interpret data from multiplexed assays of variant effect. *Genome Biology*, 20(1):223, 2019. doi: 10.1186/s13059-019-1845-6.

Boyang Fu, George Dasoulas, Sameer Gabbita, Xiang Lin, Shanghua Gao, Xiaorui Su, Soumya Ghosh, and Marinka Zitnik. STRAND: Sequence-conditioned transport for single-cell perturbations, 2026. URL https://arxiv.org/abs/2602.10156.

Fayrouz Hammal, Pierre de Langen, Aurélie Bergon, Fabrice Lopez, and Benoit Ballester. ReMap 2022: a database of human, mouse, drosophila and arabidopsis regulatory regions from an integrative analysis of DNA-binding sequencing experiments. *Nucleic Acids Research*, 50(D1): D316–D325, 2022. doi: 10.1093/nar/gkab996.

John J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982. doi: 10.1073/pnas.79.8.2554.

Julia Joung, Sai Ma, Tristan Tay, Kathryn R. Geiger-Schuller, Paul C. Kirchgatterer, Vanessa K. Verdine, Baolin Guo, Mario A. Arias-Garcia, William E. Allen, Ankita Singh, Olena Kuksenko, Omar O. Abudayyeh, Jonathan S. Gootenberg, Zhanyan Fu, Rhiannon K. Macrae, Jason D. Buenrostro, Aviv Regev, and Feng Zhang. A transcription factor atlas of directed differentiation. *Cell*, 186(1):209–229.e26, 2023. doi: 10.1016/j.cell.2022.11.026.

Kenji Kamimoto, Blerta Stringa, Christy M. Hoffmann, Kunal Jindal, Lilianna Solnica-Krezel, and Samantha A. Morris. Dissecting cell identity via network inference and in silico gene perturbation. *Nature*, 614(7949):742–751, 2023. doi: 10.1038/s41586-022-05688-9.

Konrad J. Karczewski, Laurent C. Francioli, Grace Tiao, Beryl B. Cummings, Jessica Alföldi, Qingbo Wang, Ryan L. Collins, Kristen M. Laricchia, Andrea Ganna, Daniel P. Birnbaum, et al. The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*, 581 (7809):434–443, 2020. doi: 10.1038/s41586-020-2308-7.

Dmitry Krotov and John J. Hopfield. Dense associative memory for pattern recognition. In *Advances in Neural Information Processing Systems*, volume 29, pages 1172–1180, 2016. URL https://proceedings.neurips.cc/paper/2016/hash/eaae339c4d89fc102edd9dbdb6a28915-Abstract.html.

Melissa J. Landrum, Shanmuga Chitipiralla, Kuljeet Kaur, Garth Brown, Chao Chen, Jennifer Hart, Douglas Hoffman, Wonhee Jang, Chunlei Liu, Zenith Maddipatla, Rama Maiti, Joseph Mitchell, Tayebeh Rezaie, George Riley, Guangfeng Song, Jinpeng Yang, Lora Ziyabari, Andrew Russette, and Brandi L. Kattman. ClinVar: updates to support classifications of both germline and somatic variants. *Nucleic Acids Research*, 53(D1):D1313–D1321, 2025. doi: 10.1093/nar/gkae1090.

Guole Liu, Yongbing Zhao, Yingying Zhao, Tianyu Wang, Quanyou Cai, Xiaotao Wang, Ziyi Wen, Lihui Lin, Ge Yang, and Jiekai Chen. SCARF: Single cell ATAC-seq and RNA-seq foundation model. *bioRxiv*, 2025. doi: 10.1101/2025.04.07.647689.

Ping Liu, Lyuwei Wang, Shreya Basnet, and Jianlin Cheng. TFBindFormer: A cross-attention transformer for transcription factor–DNA binding prediction. *bioRxiv*, 2026. doi: 10.64898/2026.04.09.717563.

Jaina Mistry, Sara Chuguransky, Lowri Williams, Matloob Qureshi, Gustavo A. Salazar, Erik L. L. Sonnhammer, Silvio C. E. Tosatto, Lisanna Paladin, Shriya Raj, Lorna J. Richardson, Robert D. Finn, and Alex Bateman. Pfam: The protein families database in 2021. *Nucleic Acids Research*, 49(D1):D412–D419, 2021. doi: 10.1093/nar/gkaa913.

Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Lukas Gruber, Markus Holzleitner, Thomas Adler, David P. Kreil, Michael K. Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=tL89RnzIiCd.

Ieva Rauluseviciute, Rafael Riudavets-Puig, Romain Blanc-Mathieu, Jaime A. Castro-Mondragon, Katalin Ferenc, Vipin Kumar, Roza Berhanu Lemma, Jérémy Lucas, Jeanne Chèneby, Damir Baranasic, Aziz Khan, Oriol Fornes, Sveinung Gundersen, Morten Johansen, Eivind Hovig, Boris Lenhard, Albin Sandelin, Wyeth W. Wasserman, François Parcy, and Anthony Mathelier. JASPAR 2024: 20th anniversary of the open-access database of transcription factor binding profiles. *Nucleic Acids Research*, 52(D1):D174–D182, 2024. doi: 10.1093/nar/gkad1059.

Yusuf Roohani, Kexin Huang, and Jure Leskovec. Predicting transcriptional outcomes of novel multigene perturbations with GEARS. *Nature Biotechnology*, 42(6):927–935, 2024. doi: 10.1038/s41587-023-01905-6.

Alexander Tong, Kilian Fatras, Nikolay Malkin, Guillaume Huguet, Yanlei Zhang, Jarrid Rector-Brooks, Guy Wolf, and Yoshua Bengio. Improving and generalizing flow-based generative models with minibatch optimal transport. *Transactions on Machine Learning Research*, 2024a. URL https://openreview.net/forum?id=CD9Snc73AW.

Alexander Tong, Nikolay Malkin, Kilian Fatras, Lazar Atanackovic, Yanlei Zhang, Guillaume Huguet, Guy Wolf, and Yoshua Bengio. Simulation-free schrödinger bridges via score and flow matching. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pages 1279–1287. PMLR, 2024b. URL https://proceedings.mlr.press/v238/y-tong24a.html.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017. URL https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

Michael Widrich, Bernhard Schäfl, Milena Pavlović, Hubert Ramsauer, Lukas Gruber, Markus Holzleitner, Johannes Brandstetter, Geir Kjetil Sandve, Victor Greiff, Sepp Hochreiter, and Günter Klambauer. Modern hopfield networks and attention for immune repertoire classification. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL https://proceedings.neurips.cc/paper/2020/hash/da4902cb0bc38210839714ebdcf0efc3-Abstract.html.

Wenwu Zeng, Yutao Dou, Liangrui Pan, Liwen Xu, and Shaoliang Peng. Improving prediction performance of general protein language model by domain-adaptive pretraining on DNA-binding protein. *Nature Communications*, 15(1):7838, 2024. doi: 10.1038/s41467-024-52293-7.